Nishant Kumar Singh

Backend Engineer

TypeScript · Node.js · Python · PostgreSQL

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TRY TO BREAK MY BACKENDS

SUMMARY

Backend engineer focused on the parts of a system that are expensive to get wrong: **concurrent money movement, idempotent payment APIs, exact money arithmetic, tamper-evident audit logs and approval workflows**. Failure cases are covered by tests, every project below runs CI on each push, and my portfolio lets you try to break these guarantees in five live labs.

SELECTED PROJECTS

code links point to the exact lines

Bank Management System

TypeScript · Express 5 · PostgreSQL · Prisma · Docker

Retail banking REST API: accounts, deposits, withdrawals and transfers.

- Locked every account a request touches with **SELECT ... FOR UPDATE in id order**, so racing transfers queue instead of losing updates or deadlocking; a test fires **10 parallel withdrawals** of 30.00 at 100.00 and asserts exactly 3 succeed. ↗ CODE ↗ TEST
- Built **idempotent payments**: an `Idempotency-Key` replays the original result on retry, a reused key with a new body returns 422, and a unique index settles racing duplicates. ↗ CODE
- Money as **BigInt minor units** in a double-entry ledger whose credits minus debits must equal total balances, with `CHECK` constraints as a last line of defence. ↗ CODE ↗ CODE
- Refresh tokens stored only as SHA-256 hashes; replaying a rotated token **revokes the whole token family**. ↗ CODE

Ledgerline

React · TypeScript · Hono · Cloudflare D1 · Zod

Reconciles bank statements against ledger records without silently signing off on differences.

- Matching that **refuses to guess**: tied or conflicting candidates stay unresolved for a human; money is exact integer minor units per currency. ↗ CODE
- **Independent review** (a run's creator cannot approve it) and a **hash-linked audit log** with chain verification, missing-tail detection and SQL immutability triggers; 150+ tests in CI. ↗ CODE ↗ CODE

Aftermark

TypeScript · Fastify · Bun · Cloudflare D1 · OpenAPI

Evidence-driven incident response API.

- Closing an incident needs **two non-creator approvals of the current SHA-256 evidence manifest**; new evidence makes earlier approvals stale. ↗ CODE ↗ CODE
- Optimistic concurrency: each write claims the next revision and a conflicting writer's whole batch rolls back; SQL triggers keep audit events append-only. ↗ CODE ↗ CODE

Tracebench

Python · DuckDB · Pydantic · Typer · pytest

Local-first dataset contracts and drift reports for CSV, JSONL and Parquet.

- Drift checks report **"incomparable"** instead of a false "stable" when the data cannot support a verdict; bounded ingestion, fingerprinted reports, 100+ tests. ↗ CODE

Hospital Management API

TypeScript · Express · Prisma · PostgreSQL · Redis

Hospital operations backend with payments and role-based access.

- Razorpay payments verified by **HMAC-SHA256 signature**; TOTP two-factor auth, a Redis token blacklist and role-based access. ↗ CODE

OPEN SOURCE

merged upstream

- **foss42/apidash** #1828 / Kept disabled environment variables from being lost after pre-request scripts ran.
- **CrossGL/crosstl** #3612 / Aligned HIP `__constant__` binding numbering with the CUDA importer.

SKILLS

LANGUAGES	TypeScript, JavaScript, Python, SQL	BACKEND	Node.js, Express 5, Fastify, Hono, Bun, REST, OpenAPI, Zod, Pydantic
DATA	PostgreSQL, Prisma, SQLite / Cloudflare D1, Redis, DuckDB, MySQL, MongoDB	RELIABILITY	Row locking, optimistic concurrency, idempotency keys, double-entry ledgers, hash-chained audit logs
SECURITY	JWT with refresh rotation, RBAC, HMAC signature checks, TOTP 2FA, rate limiting, input validation	DELIVERY	Vitest, pytest, Playwright, GitHub Actions, Docker, Cloudflare

Portfolio: 5 live labs, unit and browser tests run on every deploy. nishant-kumar-singh.vercel.app